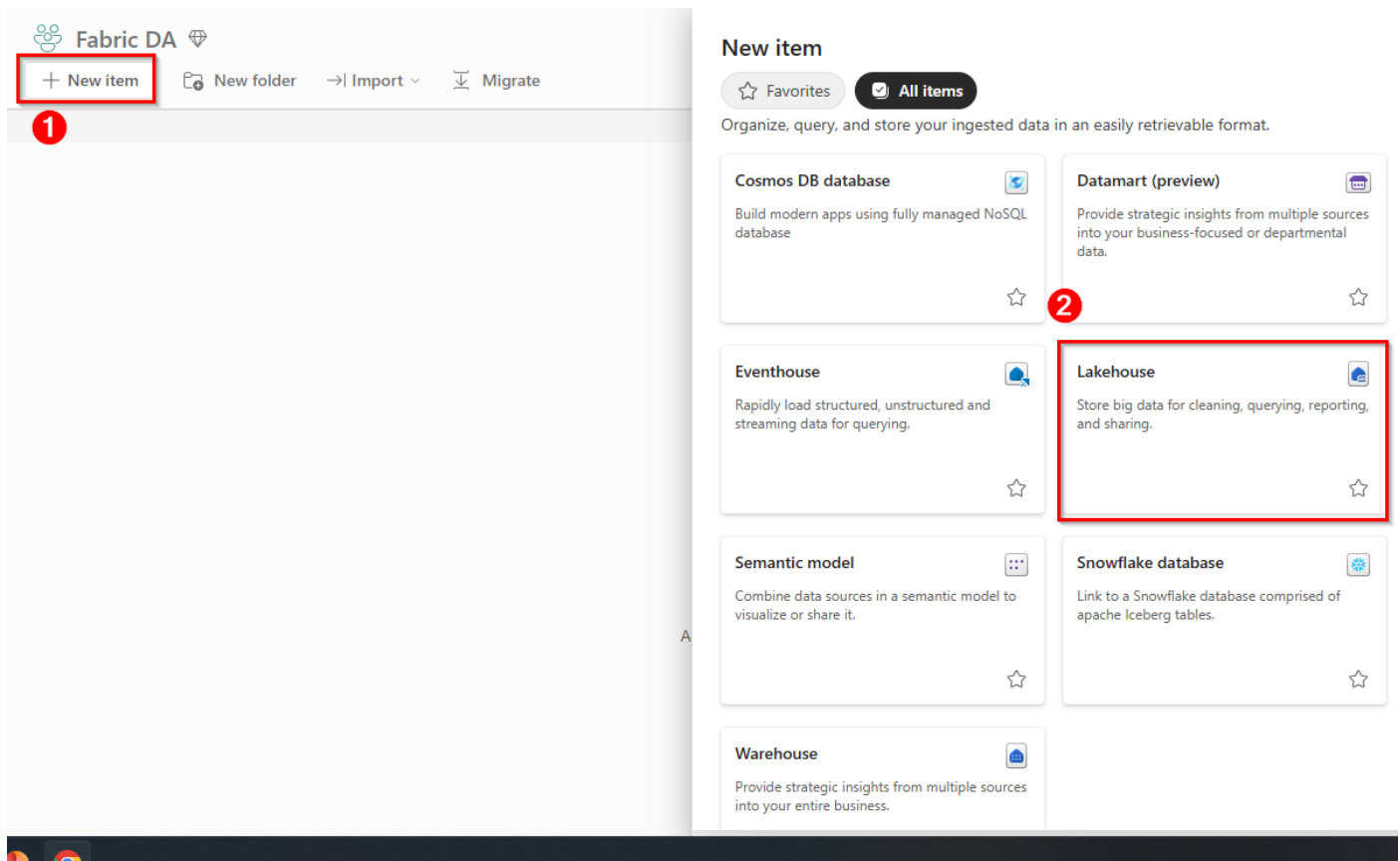


Build sales analytics with AdventureWorks dataset using Fabric Data Agent

Task 1: Create Lakehouse

1. In Fabric portal home page, Select + New item and search for Lakehouse and select it



2. Enter the Lakehouse name as AdventureWorksLH, unselect the Lakehouse schema and click create.

New Lakehouse

Name *

AdventureWorksLH

Location *

Fabric DA

☐

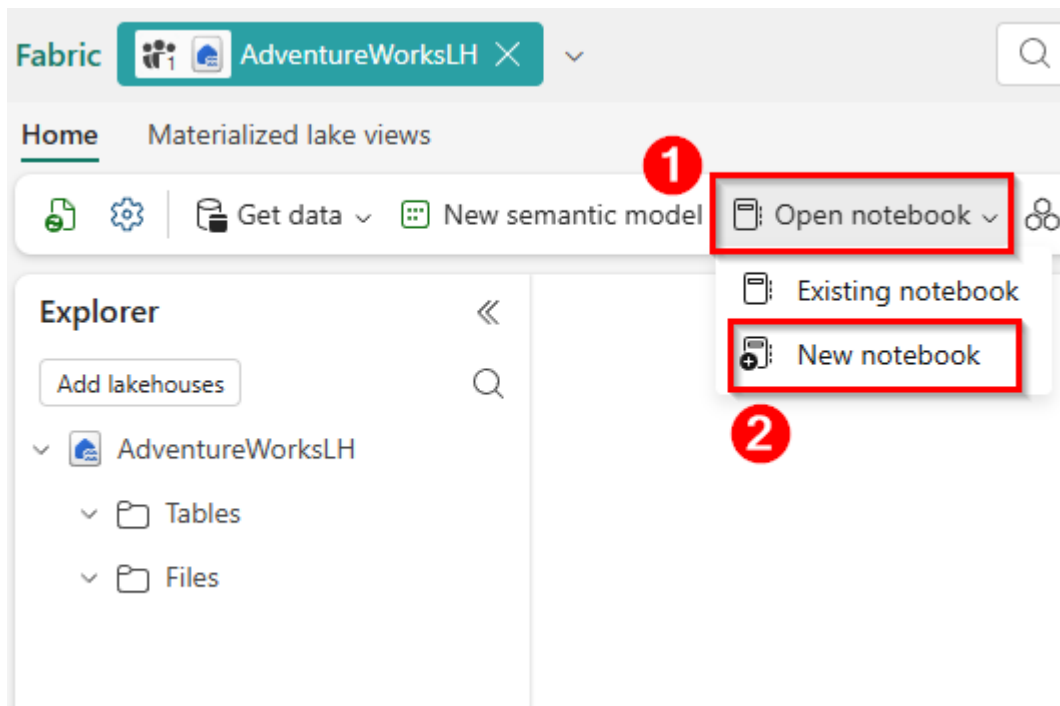
Lakehouse schemas ⓘ

Create

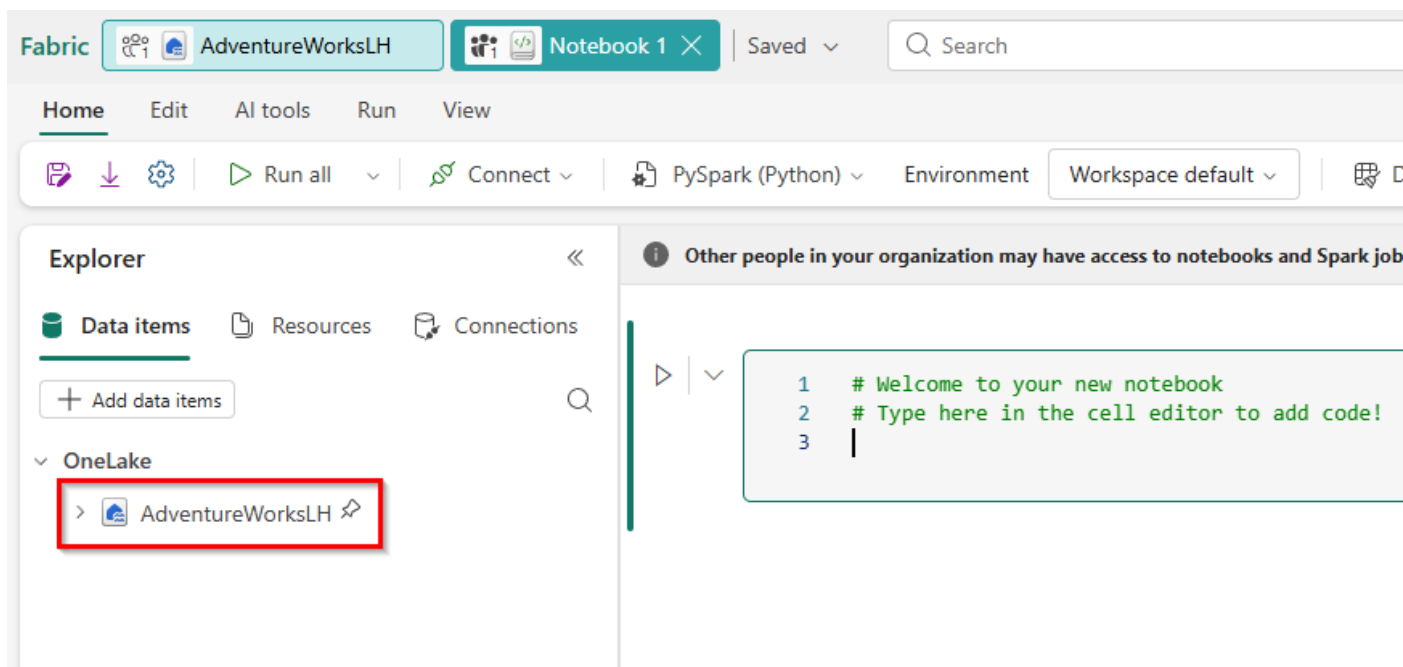
Cancel

Task 2: Populate the lakehouse with the data

1. From AdventureWorkLH, Select New notebook. This will create a new notebook and add AdventureWorkLH reference to it.



2. On the left side of the **Explorer** pane, you can see AdventureworkLH lakehouse is added.



3. In the top cell, replace all with the following code snippet:

```

import pandas as pd
from tqdm.auto import tqdm
base = "https://synapseaisolutionsa.z13.web.core.windows.net/data/AdventureWorks"

# load list of tables
df_tables = pd.read_csv(f"{base}/adventureworks.csv", names=["table"])

for table in (pbar := tqdm(df_tables['table'].values)):
    pbar.set_description(f"Uploading {table} to lakehouse")

# download
df = pd.read_parquet(f"{base}/{table}.parquet")

# save as lakehouse table
spark.createDataFrame(df).write.mode('overwrite').saveAsTable(table)

```

4. Select **Run all**.

The screenshot shows the Microsoft Fabric notebook interface. In the top toolbar, the 'Run all' button is highlighted with a red box and a red circle with the number 2. The code editor on the right contains the same Python code as the previous block, with a red box and a red circle with the number 1 highlighting the code. Below the code, a status bar shows 'Spark jobs (20 of 20 succeeded)' and a progress bar for 'Uploading FactSalesQuota to lakehouse: 100%'.

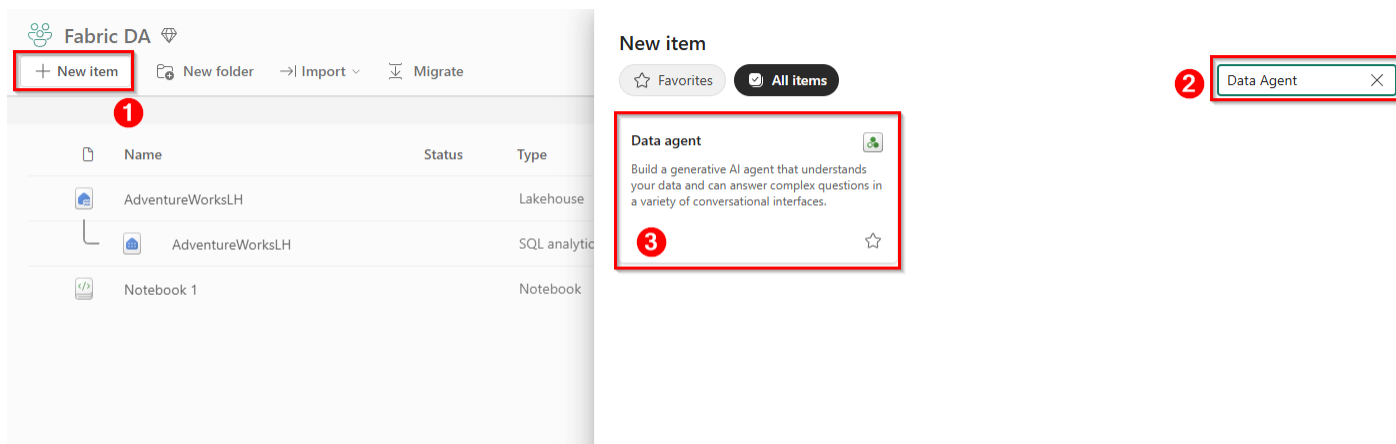
After 8-10 minutes, the lakehouse populates with the necessary data.

5. Stop the Session.

The screenshot shows the Microsoft Fabric notebook interface. The 'Standard session' button is highlighted with a red box. The interface shows the 'Home' tab selected, and the 'Standard session' button is highlighted with a red box.

Task 3: Create a Fabric data agent

1. To create a new Fabric data agent, navigate to your workspace and select the **+ New Item** button, In the All-items tab, search for Fabric data agent to locate the appropriate option as shown in this screenshot:



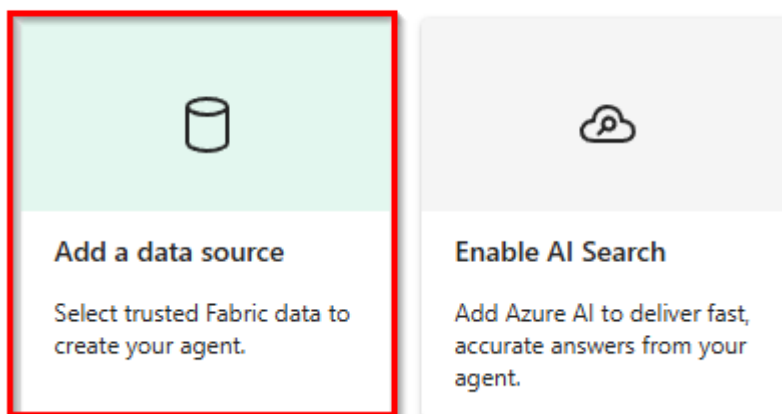
2. Once selected, a prompt asks you to provide a name for your Fabric data agent, as shown in this screenshot:

Create data agent

After you enter the name, proceed with the following steps to align the Fabric data agent with your specific requirements.

3. On Fabric Data Agent page, Select Add a data source.

Build your data agent



4. Select the AdventureWorkLH lakehouse you created in the previous step, and then select **Add**, as shown in the following screenshot:

Add a data source

All

My data

Endorsed in your org

Favorites

Filter

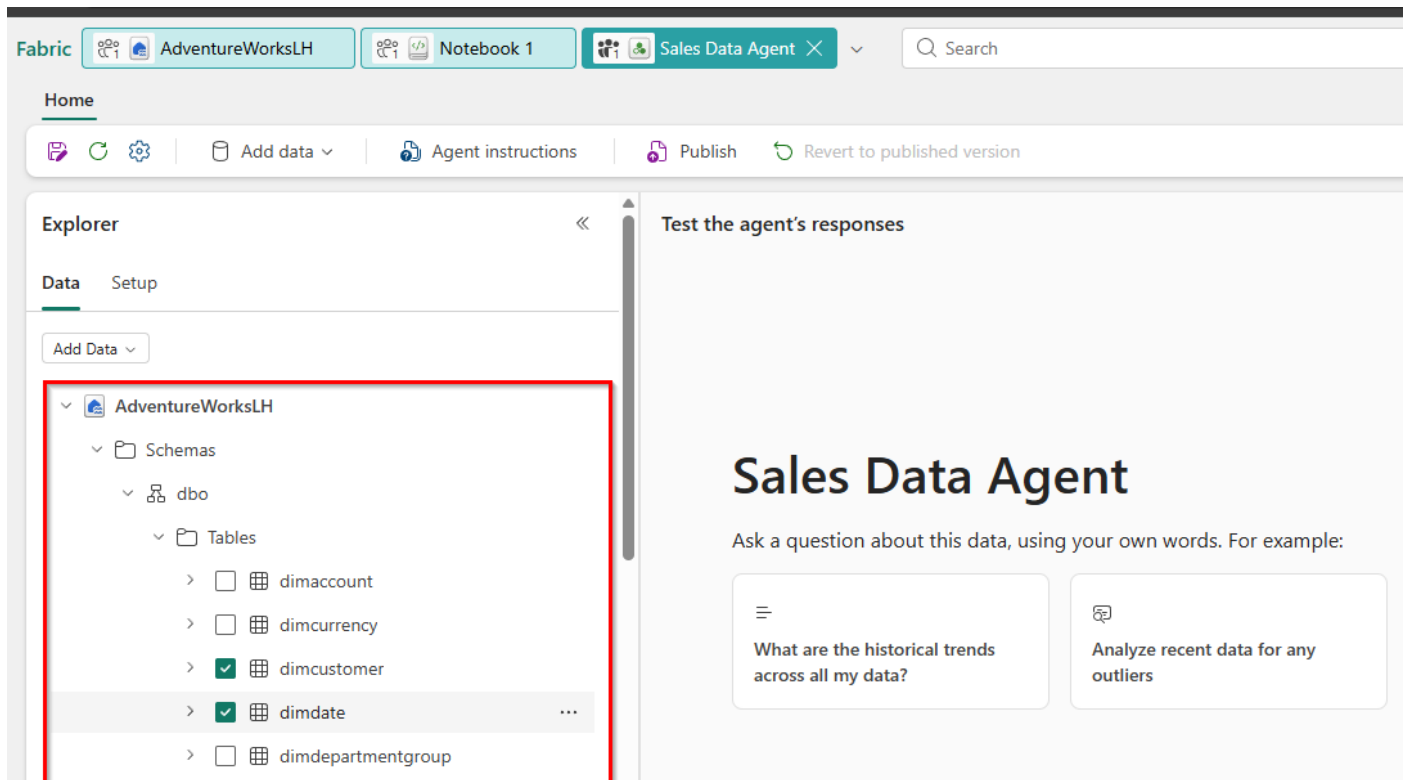
	Name	Type	Owner	Refreshed	Endorsement	Sensitivity
Explorer >>	AdventureWorksLH	Lakehouse			—	—
	Gaming_LH	Lakehouse			—	—
	Gaming_SM	Semantic model			—	—
	Gaming_EH	KQL Database			—	—
	Sales_DW	Warehouse			—	—
	WebEvents_EH	KQL Database			—	—
	Fabric Capacity Metrics	Semantic model			—	—
	DTBTutorial_EH	KQL Database			—	—
	BusModeldtm	Lakehouse			—	—
	DTBTutorial_LH	Lakehouse			—	—
	WebSalesData_LH	Lakehouse			—	—

Add

Cancel

5. Once the lakehouse is added as a data source, the Explorer pane on the left side of the Fabric data agent page shows the lakehouse name. Select the lakehouse to view all available tables. Use the checkboxes to select the tables you want to make available to the AI. For this scenario, select these tables:

- dimcustomer
- dimdate
- dimgeography
- dimproduct
- dimproductcategory
- dimpromotion
- dimreseller
- dimsalesterritory
- factinternetsales
- factresellersales



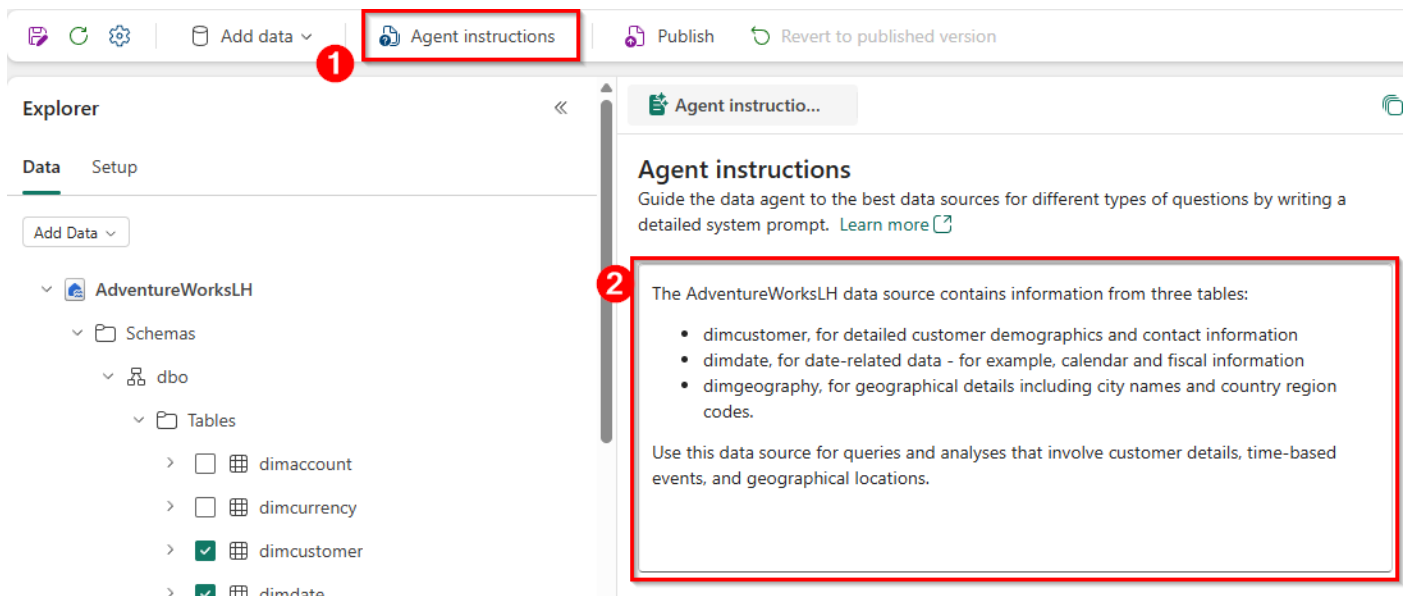
Task 4: Configure Data Agent

1. Add Data Agent instructions. To add instructions, select the **Agent instructions** button to open the instructions pane on the right. You can add the following instructions.

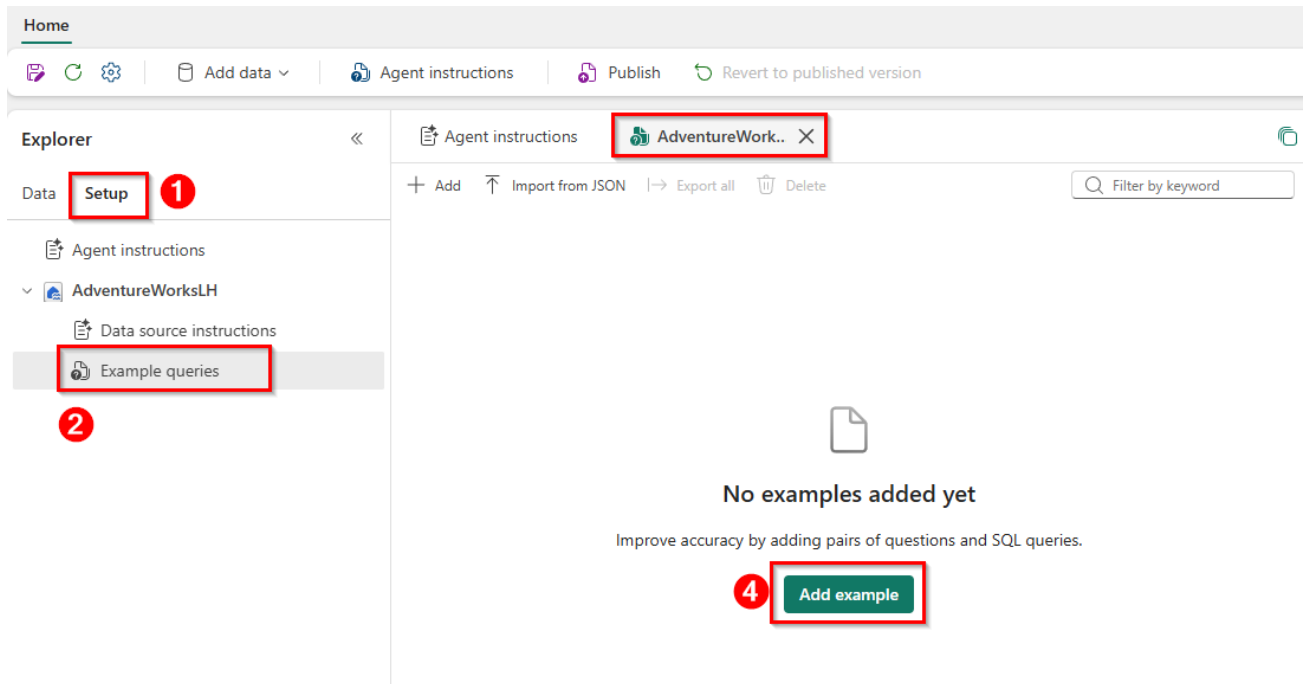
The AdventureWorksLH data source contains information from three tables:

- dimcustomer, for detailed customer demographics and contact information
- dimdate, for date-related data - for example, calendar and fiscal information
- dimgeography, for geographical details including city names and country region codes.

Use this data source for queries and analyses that involve customer details, time-based events, and geographical locations.



2. Add Examples query. To add example queries, select the **Example queries** button to open the example queries pane on the right. This pane provides options to add or edit example queries for all supported data sources. For each data source, you can select **Add or Edit Example Queries** to input the relevant examples, as shown in the following screenshot:



3. Add the following example queries for the lakehouse data source that you created. Select Add example.

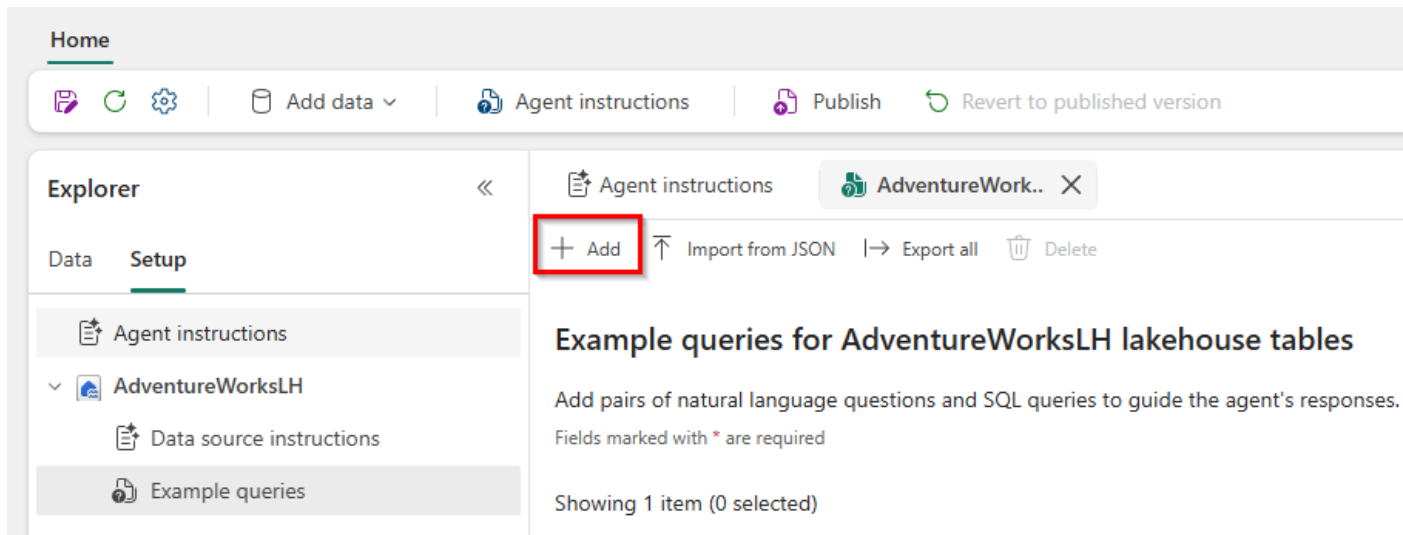
Question:

Calculate the average percentage increase in sales amount for repeat purchases for every zipcode. Repeat purchase is a purchase subsequent to the first purchase (the average should always be computed relative to the first purchase)

SQL Query:

```
SELECT AVG((s.SalesAmount - first_purchase.SalesAmount) / first_purchase.SalesAmount * 100) AS
AvgPercentageIncrease
FROM factinternetsales s
INNER JOIN dimcustomer c ON s.CustomerKey = c.CustomerKey
INNER JOIN dimgeography g ON c.GeographyKey = g.GeographyKey
INNER JOIN (
    SELECT *
    FROM (
        SELECT
            CustomerKey,
            SalesAmount,
            OrderDate,
            ROW_NUMBER() OVER (PARTITION BY CustomerKey ORDER BY OrderDate) AS RowNumber
        FROM factinternetsales
    ) AS t
    WHERE RowNumber = 1
) first_purchase ON s.CustomerKey = first_purchase.CustomerKey
WHERE s.OrderDate > first_purchase.OrderDate
GROUP BY g.PostalCode;
```

To add more query click the + Add and enter the question and SQL query.



Question:

Show the monthly total and year-to-date total sales. Order by year and month.

SQL Query:

```
SELECT
  Year,
  Month,
  MonthlySales,
  SUM(MonthlySales) OVER (PARTITION BY Year ORDER BY Year, Month ROWS BETWEEN UNBOUNDED
PRECEDING AND CURRENT ROW) AS CumulativeTotal
FROM (
  SELECT
    YEAR(OrderDate) AS Year,
    MONTH(OrderDate) AS Month,
    SUM(SalesAmount) AS MonthlySales
  FROM factinternetsales
  GROUP BY YEAR(OrderDate), MONTH(OrderDate)
) AS t
```

Now that you configured the Fabric data agent, added Fabric data agent instructions, and provided example queries for the lakehouse, you can interact with it by asking questions and receiving answers. As you continue testing, you can add more examples, and refine the instructions, to further improve the performance of the Fabric data agent. Collaborate with your colleagues to gather feedback, and based on their input, ensure the provided example queries and instructions align with the types of questions they want to ask.

Task 5: Ask questions to Fabric Data Agent

1. Customer Behavior Over Time

How many customers made purchases in each region by month?

Which cities have the highest number of new customers over time?

2. Geographic Insights

Which countries generate the highest number of customers each year?
Which countries generate the highest number of customers each year?

3. Time-Based Customer Trends

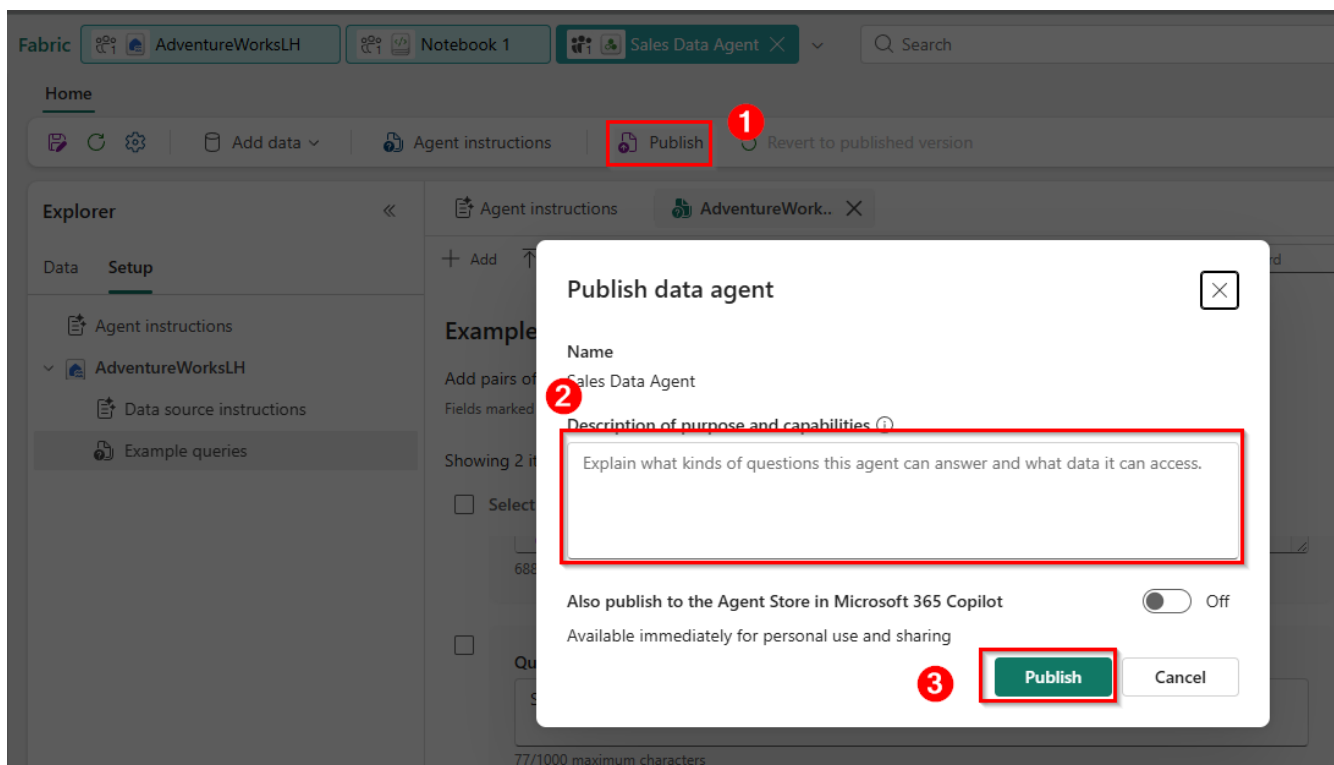
How do customer acquisitions trend over months or years in each region?
Are there any declining regions in terms of customer engagement over time?

4. Combined Insights

Which regions contribute the most revenue from repeat customers over time?
How does customer lifetime value vary across different geographies?

Task 6: Publish the Fabric data agent

After you validate the performance of the Fabric data agent, you might decide to publish it so you can then share it with your colleagues who want to do Q&A over data. In this case, select **Publish**, as shown in this screenshot:



The published URL for the Fabric data agent appears, as shown in this screenshot:

